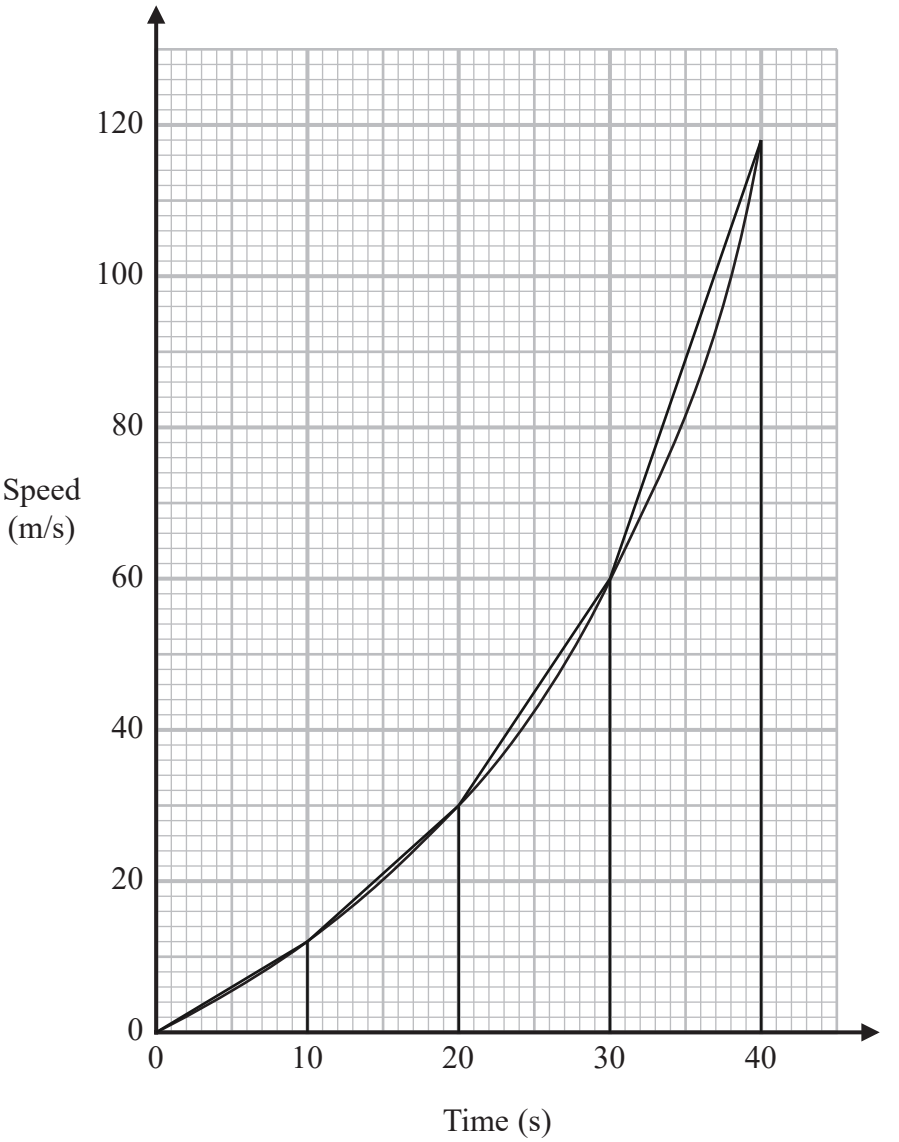
17 Here is a speed-time graph for the first 40 seconds of an aeroplane's journey.



Work out an estimate for the distance the aeroplane travelled in the first 40 seconds of its journey.
Use 4 strips of equal width.

1 mark
Use trapezium areas.

Strip width = 40 ÷ 4 = 10 s

Speeds at 0, 10, 20, 30, 40 s: 0, 12, 30, 60, 118 m/s

Distance ≈ ½ × 10 × [(0 + 118) + 2(12 + 30 + 60)]

1 mark
Include all four strips.

= 5 × 322 = 1610 m

1 mark
1610 m.

..... m

(Total for Question 17 is 3 marks)

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